

QTM 220

INTRODUCTION TO FORMAL CAUSAL INFERENCE WITH  
MULTIPLE LINEAR REGRESSION

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# READING

These notes correspond to the material from Ch. 2.2-2.3 of Hernan and Robins' *What If?*, and Ch. 3.7e, 4.7, 7.6 of Wooldridge's *Introductory Econometrics*.

# AGENDA

- ▶ Multiple linear regression with randomized experiments
- ▶ Conditionally randomized experiments and standardization
- ▶ MLR.4 and SLR.4 with conditionally randomized experiments and constant effects
- ▶ General estimation of ATE with regression and nonconstant effects

# Multiple Linear Regression with a Randomized Experiment

# NOTATION AND READING

For consistency, I'm using the notation from Wooldridge 3.7e and 7.6 in this section of the notes. There is no required reading corresponding to this material, but those interested should consult Ch. 4 of Gerber and Green *Field Experiments*.

# MULTIPLE LINEAR REGRESSION WITH A RANDOMIZED EXPERIMENT

- We showed last week that the slope from a simple linear regression (SLR) of the outcome on the treatment variable will produce an unbiased estimate of the average treatment effect (ATE) with randomized treatment assignment.

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- ▶ With randomized treatment assignment, the slope for the treatment variable in a multiple linear regression (MLR) still estimates the ATE.
- ▶ Because of this, we often use MLR to include covariates when analyzing a randomized experiment to improve efficiency.



# RECALL: RANDOMIZATION IMPLIES INDEPENDENCE

Using Wooldridge notation, suppose six subjects were randomly assigned to  $w = 0$  and  $w = 1$  (also imagine these six subjects actually represent 600 subjects), and we measure a “pre-treatment” covariate  $x$ .

| $i$ | $x_i$ | $w_i$ | $y_i$ | $y_i(1)$ | $y_i(0)$ | $y_i(1) - y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 1     | 80    | 80       | ?        | ?                 |
| 3   | 1     | 0     | 60    | ?        | 60       | ?                 |
| 4   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 6   | 1     | 0     | 80    | ?        | 80       | ?                 |

Randomization implies that  $x$  is independent of  $w$ , which implies that  $E[x|w = 1] = E[x|w = 0]$ .

# WHY INDEPENDENCE IN LARGE SAMPLES?

| $i$ | $x_i$ | $w_i$ | $y_i$ | $y_i(1)$ | $y_i(0)$ | $y_i(1) - y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 1     | 80    | 80       | ?        | ?                 |
| 3   | 1     | 0     | 60    | ?        | 60       | ?                 |
| 4   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 6   | 1     | 0     | 80    | ?        | 80       | ?                 |

- ▶ The  $w = 1$  rows are a random sample of the rows, therefore  $E[x|w = 1] = E[x]$ .
- ▶ The  $w = 0$  rows are a random sample of the rows, therefore  $E[x|w = 0] = E[x]$ .
- ▶ Therefore  $E[x|w = 1] = E[x|w = 0]$ .

# THE SOCIAL PRESSURE EXPERIMENT

## **Social Pressure and Voter Turnout: Evidence from a Large-Scale Field Experiment**

ALAN S. GERBER *Yale University*

DONALD P. GREEN *Yale University*

CHRISTOPHER W. LARIMER *University of Northern Iowa*

**V**oter turnout theories based on rational self-interested behavior generally fail to predict significant turnout unless they account for the utility that citizens receive from performing their civic duty. We distinguish between two aspects of this type of utility, intrinsic satisfaction from behaving in accordance with a norm and extrinsic incentives to comply, and test the effects of priming intrinsic motives and applying varying degrees of extrinsic pressure. A large-scale field experiment involving several hundred thousand registered voters used a series of mailings to gauge these effects. Substantially higher turnout was observed among those who received mailings promising to publicize their turnout to their household or their neighbors. These findings demonstrate the profound importance of social pressure as an inducement to political participation.

# THE NEIGHBORS TREATMENT

Dear Registered Voter:

## WHAT IF YOUR NEIGHBORS KNEW WHETHER YOU VOTED?

Why do so many people fail to vote? We've been talking about the problem for years, but it only seems to get worse. This year, we're taking a new approach. We're sending this mailing to you and your neighbors to publicize who does and does not vote.

The chart shows the names of some of your neighbors, showing which have voted in the past. After the August 8 election, we intend to mail an updated chart. You and your neighbors will all know who voted and who did not.

## DO YOUR CIVIC DUTY — VOTE!

---

| MAPLE DR                   | Aug 04 | Nov 04 | Aug 06 |
|----------------------------|--------|--------|--------|
| 9995 JOSEPH JAMES SMITH    | Voted  | Voted  | _____  |
| 9995 JENNIFER KAY SMITH    |        | Voted  | _____  |
| 9997 RICHARD B JACKSON     |        | Voted  | _____  |
| 9999 KATHY MARIE JACKSON   |        | Voted  | _____  |
| 9999 BRIAN JOSEPH JACKSON  |        | Voted  | _____  |
| 9991 JENNIFER KAY THOMPSON |        | Voted  | _____  |
| 9991 BOB R THOMPSON        |        | Voted  | _____  |
| 9993 BILL S SMITH          |        |        | _____  |
| 9989 WILLIAM LUKE CASPER   |        | Voted  | _____  |
| 9989 JENNIFER SUE CASPER   |        | Voted  | _____  |
| 9987 MARIA S JOHNSON       | Voted  | Voted  | _____  |
| 9987 TOM JACK JOHNSON      | Voted  | Voted  | _____  |
| 9987 RICHARD TOM JOHNSON   |        | Voted  | _____  |
| 9985 ROSEMARY S SUE        |        | Voted  | _____  |
| 9985 KATHRYN L SUE         |        | Voted  | _____  |
| 9985 JAMES E SUE           |        | Voted  | _____  |

# INDEPENDENCE BETWEEN PRE-TREATMENT VARIABLES AND TREATMENTS IN THE GGL EXPERIMENT

**TABLE 1. Relationship between Treatment Group Assignment and Covariates (Household-Level Data)**

|                | Control | Civic Duty | Hawthorne | Self   | Neighbors |
|----------------|---------|------------|-----------|--------|-----------|
|                | Mean    | Mean       | Mean      | Mean   | Mean      |
| Household size | 1.91    | 1.91       | 1.91      | 1.91   | 1.91      |
| Nov 2002       | .83     | .84        | .84       | .84    | .84       |
| Nov 2000       | .87     | .87        | .87       | .86    | .87       |
| Aug 2004       | .42     | .42        | .42       | .42    | .42       |
| Aug 2002       | .41     | .41        | .41       | .41    | .41       |
| Aug 2000       | .26     | .27        | .26       | .26    | .26       |
| Female         | .50     | .50        | .50       | .50    | .50       |
| Age (in years) | 51.98   | 51.85      | 51.87     | 51.91  | 52.01     |
| N =            | 99,999  | 20,001     | 20,002    | 20,000 | 20,000    |

*Note:* Only registered voters who voted in November 2004 were selected for our sample. Although not included in the table, there were no significant differences between treatment group assignment and covariates measuring race and ethnicity.

# INCLUDING COVARIATES IN THE GGL EXPERIMENT

**TABLE 3. OLS Regression Estimates of the Effects of Four Mail Treatments on Voter Turnout in the August 2006 Primary Election**

|                                                       | Model Specifications |              |              |
|-------------------------------------------------------|----------------------|--------------|--------------|
|                                                       | (a)                  | (b)          | (c)          |
| Civic Duty Treatment (Robust cluster standard errors) | .018* (.003)         | .018* (.003) | .018* (.003) |
| Hawthorne Treatment (Robust cluster standard errors)  | .026* (.003)         | .026* (.003) | .025* (.003) |
| Self-Treatment (Robust cluster standard errors)       | .049* (.003)         | .049* (.003) | .048* (.003) |
| Neighbors Treatment (Robust cluster standard errors)  | .081* (.003)         | .082* (.003) | .081* (.003) |
| N of individuals                                      | 344,084              | 344,084      | 344,084      |
| Covariates**                                          | No                   | No           | Yes          |
| Block-level fixed effects                             | No                   | Yes          | Yes          |

*Note:* Blocks refer to clusters of neighboring voters within which random assignment occurred. Robust cluster standard errors account for the clustering of individuals within household, which was the unit of random assignment.

\*  $p < .001$ .

\*\* Covariates are dummy variables for voting in general elections in November 2002 and 2000, primary elections in August 2004, 2002, and 2000.

## WARNING: INDEPENDENCE ONLY FOR PRE-TREATMENT $x$

- ▶ If  $x$  is affected by  $w$ , then the previous results will not hold.
- ▶ Suppose  $y = 1$  indicates a heart attack,  $w = 1$  is a blood pressure medicine, and  $x$  is post-treatment blood pressure.  $x$  will be affected by  $w$  and will therefore not be independent of  $w$

## SLR vs MLR WITH RANDOMIZED $w$

- ▶ In very large samples, MLR will produce approximately the same answer as SLR for the ATE (e.g., social pressure).



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- ▶ In small samples, the bias in multiple linear regression will be too much.
- ▶ The  $x$  variables can include polynomials and interaction terms, but if  $w$  is interacted with any  $x$ , then  $\beta_w$  will no longer estimate ATE.

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- ▶ In medium/large samples, (roughly  $> 30$  d.f.), multiple linear regression will have small bias but will generally be more efficient than simple linear regression.
- ▶ In small samples, the bias in multiple linear regression will be too much.
- ▶ The  $x$  variables can include polynomials and interaction terms, but if  $w$  is interacted with any  $x$ , then  $\beta_w$  will no longer estimate ATE.
- ▶ This result holds for linear models used with limited dependent variables (LDV) such as the linear probability model when used with binary  $y$ , but robust SEs must be used.

# INTERACTIONS AND LDVs

- ▶ There can be small sample advantages to interacting  $w$  with the  $x$  variables but the APE should be used to estimate the ATE (see Ch. 6-2d on Wooldridge pp. 194-195).

# INTERACTIONS AND LDVs

- ▶ There can be small sample advantages to interacting  $w$  with the  $x$  variables but the APE should be used to estimate the ATE (see Ch. 6-2d on Wooldridge pp. 194-195).
- ▶ There can be small sample advantages to using LDV models such as logit, but the APE should be used to estimate the ATE (e.g., see formula 17.17 on Wooldridge pg. 567 for the logit model).

# Conditionally Randomized Experiments

# NOTATION AND READING

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# CONDITIONALLY RANDOMIZED EXPERIMENT

Suppose for one group/block/stratum ( $x = 1$ ) a coin was flipped to assign treatment to half of the units, and for another group/block stratum ( $x = 0$ ), one out of four units was assigned to treatment. This is called a blocked, stratified, or conditionally randomized experiment. Essentially, we now have two experiments.

| $i$ | $x_i$ | $w_i$ | $y_i$ | $y_i(1)$ | $y_i(0)$ | $y_i(1) - y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 1     | 80    | 80       | ?        | ?                 |
| 3   | 0     | 0     | 60    | ?        | 60       | ?                 |
| 4   | 1     | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 6   | 0     | 0     | 80    | ?        | 80       | ?                 |

# CONDITIONALLY RANDOMIZED EXPERIMENT

Within each of our two groups ( $x = 0$  and  $x = 1$ ), all the properties discussed last week hold. Within  $x = 0$ ,

| $i$ | $x_i$ | $w_i$ | $y_i$ | $y_i(1)$ | $y_i(0)$ | $y_i(1) - y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

remember red numbers not observed

What is the average  $y_i(1)$  among the  $x = 0$  subjects?

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| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

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What is the average  $y_i(1)$  among the  $x = 0$  subjects?

$$\frac{70 + 60 + 70 + 80}{4} = 70$$

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| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

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What is the average  $y_i(1)$  among the  $x = 0, w = 1$  subjects?

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|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

remember red numbers not observed

What is the average  $y_i(1)$  among the  $x = 0, w = 1$  subjects?

$$\frac{70}{1} = 70$$

# CONDITIONALLY RANDOMIZED EXPERIMENT

Within each of our two groups ( $x = 0$  and  $x = 1$ ), all the properties discussed last week hold. Within  $x = 0$ ,

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|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

remember red numbers not observed

What is the average  $y_i(0)$  among the  $x = 0$  subjects?

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Within each of our two groups ( $x = 0$  and  $x = 1$ ), all the properties discussed last week hold. Within  $x = 0$ ,

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|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

remember red numbers not observed

What is the average  $y_i(0)$  among the  $x = 0$  subjects?

$$\frac{70 + 60 + 70 + 80}{4} = 70$$

# CONDITIONALLY RANDOMIZED EXPERIMENT

Within each of our two groups ( $x = 0$  and  $x = 1$ ), all the properties discussed last week hold. Within  $x = 0$ ,

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|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

remember red numbers not observed

What is the average  $y_i(0)$  among the  $x = 0, w = 0$  subjects?



# CONDITIONALLY RANDOMIZED EXPERIMENT

Within each of our two groups ( $x = 0$  and  $x = 1$ ), all the properties discussed last week hold. Within  $x = 0$ ,

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|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

remember red numbers not observed

What is the average  $y_i(0)$  among the  $x = 0, w = 0$  subjects?

$$\frac{60 + 70 + 80}{3} = 70$$

## CONDITIONAL ATE FOR $x = 0$

Because we have a randomized experiment within the  $x = 0$  group and because of the linearity of expectations, the SLR slope (difference in means) among the  $x = 0$  group provides an unbiased estimator of the conditional average treatment effect ( $CATE_{x=0}$ ).

$$\begin{aligned} & \frac{70}{1} - \frac{60 + 70 + 80}{3} \\ &= \frac{70 + 60 + 70 + 80}{4} - \frac{70 + 60 + 70 + 80}{4} \\ &= \frac{0 + 0 + 0 + 0}{4} \end{aligned}$$

# CONDITIONALLY RANDOMIZED EXPERIMENT

Within the  $x = 1$  group, all the properties discussed last week hold.

| $i$ | $x_i$ | $w_i$ | $y_i$ | $y_i(1)$ | $y_i(0)$ | $y_i(1) - y_i(0)$ |
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| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
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| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $y_i(1)$  among the  $x = 1$  subjects?

$$\frac{80 + 80}{2} = 80$$

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| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
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| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $y_i(1)$  among the  $x = 1, w = 1$  subjects?

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| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
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| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $y_i(0)$  among the  $x = 1$  subjects?

# CONDITIONALLY RANDOMIZED EXPERIMENT

Within the  $x = 1$  group, all the properties discussed last week hold.

| $i$ | $x_i$ | $w_i$ | $y_i$ | $y_i(1)$ | $y_i(0)$ | $y_i(1) - y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $y_i(0)$  among the  $x = 1$  subjects?

$$\frac{70 + 70}{2} = 70$$



# CONDITIONALLY RANDOMIZED EXPERIMENT

Within the  $x = 1$  group, all the properties discussed last week hold.

| $i$ | $x_i$ | $w_i$ | $y_i$ | $y_i(1)$ | $y_i(0)$ | $y_i(1) - y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $y_i(0)$  among the  $x = 1, w = 0$  subjects?

# CONDITIONALLY RANDOMIZED EXPERIMENT

Within the  $x = 1$  group, all the properties discussed last week hold.

| $i$ | $x_i$ | $w_i$ | $y_i$ | $y_i(1)$ | $y_i(0)$ | $y_i(1) - y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $y_i(0)$  among the  $x = 1, w = 0$  subjects?

$$\frac{70}{1} = 70$$

## CONDITIONAL ATE FOR $x = 1$

Because we have a randomized experiment within the  $x = 1$  group and because of the linearity of expectations, the SLR slope (difference in means) among the  $x = 1$  group provides an unbiased estimator of the conditional average treatment effect ( $CATE_{x=1}$ ).

$$\begin{aligned} & \frac{80}{1} - \frac{70}{1} \\ &= \frac{80 + 80}{2} - \frac{70 + 70}{2} \\ &= \frac{10 + 10}{2} \end{aligned}$$

# STANDARDIZATION

If we average the  $CATE_{x=0}$  and  $CATE_{x=1}$  by the proportion in each group/block/stratum, then we get the ATE. In epidemiology, this is known as standardization.

| $i$ | $x_i$ | $w_i$ | $y_i$ | $y_i(1)$ | $y_i(0)$ | $y_i(1) - y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | ?        | ?                 |
| 2   | 1     | 1     | 80    | 80       | ?        | ?                 |
| 3   | 0     | 0     | 60    | ?        | 60       | ?                 |
| 4   | 1     | 0     | 70    | ?        | 70       | ?                 |
| 5   | 0     | 0     | 70    | ?        | 70       | ?                 |
| 6   | 0     | 0     | 80    | ?        | 80       | ?                 |

$$\frac{4}{6} \cdot \left\{ \frac{70}{1} - \frac{60 + 70 + 80}{3} \right\} + \frac{2}{6} \cdot \left\{ \frac{80}{1} - \frac{70}{1} \right\} \\ = \frac{20}{6}$$

# STANDARDIZATION

If we average the  $CATE_{x=0}$  and  $CATE_{x=1}$  by the proportion in each group/block/stratum, then we get the ATE. In epidemiology, this is known as standardization.

| $i$ | $x_i$ | $w_i$ | $y_i$ | $y_i(1)$ | $y_i(0)$ | $y_i(1) - y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

$$\begin{aligned} \frac{4}{6} \cdot \left\{ \frac{70}{1} - \frac{60 + 70 + 80}{3} \right\} &+ \frac{2}{6} \cdot \left\{ \frac{80}{1} - \frac{70}{1} \right\} \\ &= \frac{20}{6} = \frac{10 + 10}{6} \end{aligned}$$

# OMITTED VARIABLE BIAS WITH SLR

If we ignore the groups, and run an SLR on the entire data, we get omitted variable bias.

| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1) - Y_i(0)$ ?

# OMITTED VARIABLE BIAS WITH SLR

If we ignore the groups, and run an SLR on the entire data, we get omitted variable bias.

| $i$ | $X_i$ | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ | $Y_i(1) - Y_i(0)$ |
|-----|-------|-------|-------|----------|----------|-------------------|
| 1   | 0     | 1     | 70    | 70       | 70       | 0                 |
| 2   | 1     | 1     | 80    | 80       | 70       | 10                |
| 3   | 0     | 0     | 60    | 60       | 60       | 0                 |
| 4   | 1     | 0     | 70    | 80       | 70       | 10                |
| 5   | 0     | 0     | 70    | 70       | 70       | 0                 |
| 6   | 0     | 0     | 80    | 80       | 80       | 0                 |

What is the average  $Y_i(1) - Y_i(0)$ ?

$$\begin{aligned}\frac{0 + 10 + 0 + 10 + 0 + 0}{6} &= \frac{20}{6} \\ &\approx 3.3 \\ &\neq \frac{70 + 80}{2} - \frac{60 + 70 + 70 + 80}{4}\end{aligned}$$

# SUMMARY OF CONDITIONALLY RANDOMIZED EXPERIMENTS

- ▶ A conditionally randomized experiment can be analyzed as multiple experiments and these results can be combined with standardization.
- ▶ If you ignore the groups in a conditionally randomized experiment, and the probability of treatment assignment is different in the groups, then you will induce omitted variable bias.



# UP NEXT

In the synchronous session this week, we will connect these results to material from Wooldridge 3.7e, 4.7, and 7.6. Some foreshadowing:

- ▶ Many non-experimental studies can be thought of as approximating a conditionally randomized experiment.
- ▶ Wooldridge presents techniques in 7.6 that are equivalent to standardization when  $x$  is discrete.
- ▶ In 3.7e and 4.7, Wooldridge presents models and methods for constant effects. With constant effects, a regression of  $y$  on  $w$  and  $x$  will be preferred to standardization. Without constant effects, a regression of  $y$  on  $w$  and  $x$  will produce a weighted average treatment effect.

# Synchronous Material

# RANDOM TREATMENT ASSIGNMENT AND SLR.4 WITH CONSTANT TREATMENT EFFECTS

$$\begin{aligned}y_i &= y_i(0) + te_i w_i \\&= E[y_i(0)] + \tau w_i + [y_i(0) - E[y_i(0)]] \\&= \beta_0 + \beta_1 w_i + u_i\end{aligned}$$

Randomized  $w$  implies that  $w$  is independent of  $[y(0), y(1)]$   
This implies that SLR.4 holds,

$$\begin{aligned}E[u|w] &= E[y_i(0) - E[y_i(0)]|w] \\&= E[y_i(0)|w] - E[E[y_i(0)]|w] \\&= E[y_i(0)] - E[y_i(0)] \\&= 0\end{aligned}$$

## WITH COVARIATES WITH CONSTANT EFFECTS

As discussed in the “MLR with Randomized Experiments” video, randomized  $w$  also implies that  $w$  is independent of  $[y(0), y(1)]$  conditional on  $\mathbf{x}$  (as long as  $\mathbf{x}$  is not affected by  $w$ ). Hence, with constant treatment effects,  $y = y(0) + \tau w$  and

$$\begin{aligned} E[y|w, \mathbf{x}] &= E[y(0)|w, \mathbf{x}] + E[\tau w|w, \mathbf{x}] \\ &= E[y(0)|w, \mathbf{x}] + \tau w \end{aligned}$$

Now if we assume that  $E[y(0)|\mathbf{x}] = \alpha + \mathbf{x}\gamma$  and write  $u = y(0) - \alpha + \mathbf{x}\gamma$ , then MLR.4 holds,

$$\begin{aligned} E[u|w, \mathbf{x}] &= E[y(0) - (\alpha + \mathbf{x}\gamma)|w, \mathbf{x}] \\ &= E[y(0)|w, \mathbf{x}] - (\alpha + \mathbf{x}\gamma) \\ &= E[y(0)|\mathbf{x}] - (\alpha + \mathbf{x}\gamma) \\ &= \alpha + \mathbf{x}\gamma - (\alpha + \mathbf{x}\gamma) \\ &= 0 \end{aligned}$$

# CONDITIONAL VS UNCONDITIONAL WITH CONSTANT EFFECTS

If  $w$  is randomized unconditionally (aka marginally),

- ▶ this implies that  $w$  is independent of  $[y(0), y(1)]$  and that  $w$  is independent of  $[y(0), y(1)]$  conditional on  $\mathbf{x}$  (when  $\mathbf{x}$  is not affected by  $w$ ). Hence, SLR.4 and MLR.4 hold. (Note that MLR.4 will hold for lots of different  $\mathbf{x}$ .)

If  $w$  is randomized conditionally on  $\mathbf{x}$ ,

- ▶ then  $w$  is independent of  $[y(0), y(1)]$  conditional on  $\mathbf{x}$ , but  $w$  is **not** independent of  $[y(0), y(1)]$  unconditionally (unless  $P(w = 1|\mathbf{x}) = P(w = 1)$ )
- ▶ If  $E[y(0)|\mathbf{x}]$  is linear, then MLR.4 holds, but SLR.4 does not hold, and SLR will lead to omitted variable bias.

# APPROXIMATING CONDITIONAL RANDOMIZATION

In non-experimental studies, conditioning on variables  $\mathbf{x}$  that affect both the treatment  $w$  and the outcome  $y$  will often lead to a better approximation of a conditionally randomized experiment than a simple regression of  $y$  on  $w$  will approximate an unconditionally randomized experiment.

In example 3.7 on pg. 101, we have

- ▶  $\widehat{earn98} = 10.61 - 2.05train$  with  $se_{train} = 0.48$
- ▶  $\widehat{earn98} = 4.67 + 2.41train + .373earn96 + .363educ - .181age + 2.48married$  with  $se_{train} = 0.44$

We conclude that SLR.4 likely does not hold. Whether MLR.4 holds approximately is debatable.

## WITH NONCONSTANT EFFECTS

If  $w$  is conditionally randomized on  $\mathbf{x}$  or we think that  $w$  is approximately independent of  $[y(0), y(1)]$  conditional on  $\mathbf{x}$ , then we can estimate the ATE as

$$\frac{1}{n} \sum_{i=1}^n \{\hat{y}_i(1) - \hat{y}_i(0)\}$$

For  $\hat{y}_i(1)$

1. Fit a regression of  $y$  on  $\mathbf{x}$  for just the  $w = 1$  observations.
2. Using this regression, predict  $\hat{y}_i$  using  $\mathbf{x}_i$  for all observations (i.e., with  $w_i = 1$  and  $w_i = 0$ )

For  $\hat{y}_i(0)$

1. Fit a regression of  $y$  on  $\mathbf{x}$  for just the  $w = 0$  observations.
2. Using this regression, predict  $\hat{y}_i$  using  $\mathbf{x}_i$  for all observations (i.e., with  $w_i = 1$  and  $w_i = 0$ )

Note that we haven't specified the type of regression.

On Wooldridge pg. 247, we see that with linear models, this approach will produce

$$\hat{y}_i(0) = \hat{\alpha}_0 + \hat{\gamma}_{0,1}x_{i1} + \hat{\gamma}_{0,2}x_{i2} + \cdots + \hat{\gamma}_{0,k}x_{ik}$$

$$\hat{y}_i(1) = \hat{\alpha}_1 + \hat{\gamma}_{1,1}x_{i1} + \hat{\gamma}_{1,2}x_{i2} + \cdots + \hat{\gamma}_{1,k}x_{ik}$$

with

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \{\hat{y}_i(1) - \hat{y}_i(0)\} \\ = (\hat{\alpha}_1 - \hat{\alpha}_0) + (\hat{\gamma}_{1,1} - \hat{\gamma}_{0,1})\bar{x}_1 + \cdots + (\hat{\gamma}_{1,k} - \hat{\gamma}_{0,k})\bar{x}_k \end{aligned}$$

Note that if we center the  $\mathbf{x}$  variables such that their means are zero and this formula is greatly simplified.



1. Draw a picture demonstrating this linear approach with a single  $x$ .
2. Draw a picture demonstrating this approach with nonlinearities in  $x$ .
3. Draw a picture demonstrating this approach with logistic regression.

# STANDARD ERRORS

Notice that these approaches are equivalent to the average partial effect (APE) approaches we studied previously. For standard errors,

1. Use the package (like the one we showed you for logit)
2. Bootstrap

# HOW TO CHOOSE $\mathbf{x}$

VanderWeele and Shpitser (2011), “A new criterion for confounder selection,” *Biometrics* proved that a disjunctive cause criterion was sufficient for  $w$  to be independent of  $[y(0), y(1)]$  conditional on  $\mathbf{x}$ .

Disjunctive Cause Criterion:

1. Assume there is a correct set of pre-treatment variables in our data.
2. Then a set  $\mathbf{x}$  that includes all variables that affect  $w$ ,  $y$ , or both  $w$  and  $y$ , will be sufficient for  $w$  to be independent of  $[y(0), y(1)]$  conditional on  $\mathbf{x}$ .